

ProActive Plan B

Student instructions and audit specification

Existing-cache answer selection | 16 September 2026

1. Your assignment in one paragraph

Test whether a small selector can improve a vision-language model's answer by choosing between its original answer and alternatives already stored in the ProActive cache. Start with the original answer plus the grounding and blur answers. Use existing data only. Do not generate new answers from Qwen or Gemma. This document is a handoff for the proposed experiment, not evidence that the selector has already been trained or succeeds.

2. Models and datasets

Item	Instruction
Large-model outputs	Reuse Qwen3-VL-8B and Gemma-4-E4B cached answers. Do not fine-tune or query these large models.
Small model to train	One pooled logistic-regression answer selector using the technical specification's fixed features.
Core data	Closed-answer records from POPE, HallusionBench and VSR, subject to the exact contract and image-overlap filters in the specification.
Excluded from primary study	VizWiz/free-form exact-match proxy results. They may be descriptive appendix results only, clearly labeled.
Additional evaluation	PRE-HAL and IllusionBench. These datasets have already been examined, so the correction evaluation is exploratory.
Primary evidence budget	Grounding plus blur: two additional probes, plus the existing clean answer.

HallusionBench and IllusionBench are different datasets. Use the manifest IDs, not approximate name matching. Dataset eligibility must follow the actual answer contract, not just the dataset name.

3. What the audit established

The research task inspected ten recovered cache shards at repository revision 14768af3de4436e95238274780efaadbbf8e01db. It reported matching recomputed original-answer correctness to all 16,982 cached records. These audit findings are the starting point; reproduce the manifest and filters before fitting.

Original split	Model-example pairs
Training	10,118
Validation	1,404
Calibration	1,500
Core test	1,560
Held-out dataset shift	2,400

The proposed filtered fitting cohort is 1,898 training pairs and 314 validation pairs. These are model-example pairs, not necessarily unique images. Preserve paired models and image groups in statistical analysis. Do not force counts to match by arbitrarily dropping records; investigate differences and record the exact inclusion ledger.

4. Important data problems to handle first

- 675 probe answers were unusable despite valid/applicable flags. An attempted unusable probe still costs one check, but supplies neither a candidate nor a vote.
- Confidence extraction is not verifiably repaired. Exclude confidence, entropy, margins, confidence shifts and diagnostic features derived from them. The selector's fitted probability is a different quantity; use it only as specified.
- 654 metadata image identities cross original splits. Original grouped IDs alone are insufficient protection. Apply the specification's image exclusions before fitting and save the excluded-ID list.
- No untouched evaluation cohort is certified. Describe this experiment as exploratory even if train/test groups are properly separated.
- Retained latency excludes historical failed attempts. Report additional acquired-probe count as the primary cost; do not call it a verified end-to-end runtime saving.

5. Implementation sequence

Step 1 - Reproduce the audit. Retrieve the pinned inputs and the original audit bundle. Verify hashes, fields, scoring contracts and cohort counts. Keep frozen repository outputs unchanged.

Step 2 - Freeze the experiment configuration. Record the feature list, candidate rules, image exclusions, logistic settings, switching-margin grid, baseline definitions, comparator selection, statistical procedure and go/no-go gates. The technical appendix identifies any parameters the original specification leaves unspecified. Resolve those before fitting, without using test performance.

Step 3 - Build candidates. For each model-example pair, start with clean and acquire grounding plus blur. Parse and normalize each answer using the declared contract. Handle invalid candidates and duplicate answers exactly as specified. Gold answers may generate training targets and evaluation scores but must not enter prediction features.

Step 4 - Reproduce simple baselines. Implement keep-original, always-ground, majority voting and supported-ground. Compare predictions on identical examples and evidence pools. Also report each baseline's minimum operational probe cost separately.

Step 5 - Fit the small selector. Use training data only. Select the switching margin on validation only. Follow the fixed specification rather than searching new architectures, feature sets or probe orders.

Step 6 - Freeze and evaluate. Save code revision, fitted parameters, preprocessing and validation decision before evaluating core test and shift. Do not revise the selector in response to these exploratory test results and then present the revised score as a fresh confirmation.

Step 7 - Report every slice. Produce pooled, per-dataset and per-model results with denominators. Keep closed-answer primary results separate from free-form proxies. Include negative results and unusable-answer counts.

6. What the comparisons mean

Method	Action	Minimum operational cost
Keep original	Return clean answer	0 additional probes
Always ground	Return usable grounding answer, with declared fallback	1 additional probe
Majority vote	Vote over the clean and acquired usable answers	2 probes in the primary experiment
Supported ground	Switch to grounding only when another acquired answer supports it	Up to 2 probes; use the fixed accounting rule
Logistic selector	Score candidates and switch only under the validation-frozen rule	2 probes in the primary experiment
Oracle	Choose a correct candidate if one exists, using gold	Privileged upper bound, not a deployable method

Matched-evidence analysis can charge all methods two probes because the same pool was acquired. That must not hide the fact that keep-original and always-ground can operate with fewer calls. Report both views explicitly. This experiment uses a fixed grounding-plus-blur acquisition schedule; it does not by itself prove that ProActive's learned acquisition policy improves correction.

7. Metrics the student must return

- Answer accuracy: correct selected answers divided by evaluated examples.
- Repairs: examples with a wrong original answer and a correct selected answer.
- Damage: examples with a correct original answer and a wrong selected answer.
- BreakRate: damage divided by the number of originally correct examples, not by all examples.
- Mean additional acquired probes: excludes the clean generation; includes attempted unusable probes.
- Paired improvement and its interval: compare methods on the same image-clustered examples using the appendix's procedure.

For binary correctness, final correct count must equal original correct count + repairs - damage. Check this identity in every reported slice. A high oracle score is an opportunity bound; it is not a learned-system result.

8. Existing reference numbers, not selector results

For the proposed closed-answer core test, the audit reported keep-original accuracy 85.12%, always-ground 85.23%, majority vote 85.01%, and oracle 90.59% with grounding-plus-blur available. Reproduce the exact cohort and scores before comparing a trained selector. The broader core aggregate includes free-form exact-match proxies and must not replace this primary cohort.

The proposed success gate asks for at least one percentage point improvement over a validation-frozen simple comparator, statistical support, BreakRate at most 1%, and no substantial dataset/model regression. Use the exact formal gate in the technical specification. If it fails, report the failure; do not adjust the threshold afterward to manufacture a positive conclusion.

9. Required handback package

- A short README containing exact reproduction commands and the pinned source revision.
- Input manifest and hashes; inclusion/exclusion ledger; usable-candidate audit; image-overlap report.
- Frozen configuration, code revision, fitted small-model parameters and validation selection record.
- Per-example prediction file with split, dataset, model, image/group ID, selected source, cost and evaluation outcomes. Keep evaluation-only gold fields out of the predictor interface.
- CSV tables per split/dataset/model and pooled, including N, original accuracy, final accuracy, repairs, damage, BreakRate, probe cost and paired intervals.
- A one-page conclusion: gate passed or failed, which slices help or regress, limitations and whether a further experiment is justified.

10. Scope and source access

Do not run new large-model inference, repeated sampling, verifier calls, target-specific checks, new architecture searches or model fine-tuning for this assignment. No such inference is needed for the proposed cached-data selector. This PDF does not claim that missing repeated-sampling or verifier baselines have been evaluated.

Repository: [ProActive-ICLR-Dump](#).

Audit and downloadable artifacts: [Analyze GitHub Research Findings](#). This is an account-access conversation link; the project owner should provide the actual downloaded bundle to the student. A conversation link alone does not guarantee student access.

Read the technical specification below before implementing. It takes precedence over this plain-language overview for exact definitions. Preserve any expressly unresolved settings as a pre-training decision, rather than guessing silently.

Technical specification and provenance

The following text was retrieved from the research task in response to a request for the audit specification. It is reproduced with formatting changes. Any stated omissions or unresolved settings remain unresolved; this handoff does not silently replace them.

Appendix A - Original proposed experiment specification

Source file: Proposed_Experiment_Specification.md

Original SHA-256: 4b32e10fc902622fa9425dab3410df871661c71428a5adb569d823ca098dfed2

Proposed freeze: one small answer selector

Status: PROPOSED, NOT APPROVED OR RUN. Created after descriptive cache inspection on 2026-09-16. This is an exploratory protocol freeze, not a preregistration before any outcome inspection. Training is outside the present authorization.

Question and permitted claim

Can a confidence-free small model decide whether to keep the original answer or choose an already acquired alternative, improving final-answer correctness at the same acquired-probe cost? This isolates answer selection, not learned acquisition, generation, or diagnostic-label prediction. Any positive claim is exploratory and restricted to the declared answer contracts and cached candidate distribution.

Immutable inputs and scope

Use only the ten SHA-pinned recovered caches in `reference_results/audit_summary.json`, repository commit 14768af3de4436e95238274780efaadbbf8e01db. No new language-model or embedding calls. No confidence repair, prompt alteration, teacher-label change or cache mutation.

Primary fitting/evaluation scope: closed-answer POPE, VSR and binary Hallusion rows. Exclude VizWiz and open-ended Hallusion from fitting and primary evaluation because this audit's exact-match scoring is not a satisfactory general free-form correctness contract. Continue reporting the all-contract descriptive audit; do not hide it. Shift PRE-HAL/IllusionBench is a secondary exploratory transfer evaluation, not a tuning set.

Use `cohort_index.csv` flags: train/validation require both `closed_answer_eligible=1` and `proposed_fit_eligible=1`. Proposed image purge removes train images present in any later split and validation images present in calibration/test/shift. Counts: train 1,898 pairs (949 base examples), validation 314 (157); core closed test 914 (457). Keep original calibration unused for selector choices. Shift contains 2,400 pairs. Both models and all prefixes of an instance stay together. Metadata image identities define clustering/protection; pixel-level deduplication is not certified.

Before fitting, verify hashes, eligibility counts and train/validation/evaluation image intersections. Stop on discrepancies; record a protocol amendment rather than silently repairing cohorts. This audit already computes descriptive results on all splits, so none is claimed untouched.

Acquisition and scoring

Fixed order: grounding, blur, crop, brightness, noise, blank. Primary budget $B=2$ (grounding + blur); secondary $B=1,3,5,6$. All positive-budget methods receive the identical prefix and pay its acquired logical slots. Original answer is excluded from additional cost. No learned acquisition policy is used.

Unavailable/not-applicable slot: skip, no replacement, zero cost. Acquired invalid outcome: one cost, no candidate. In the actual audited mandatory slots no unavailable/not-applicable case occurs. Recovered grounding counts as one effective slot; historic retry-inclusive cost is unknown and must be qualified.

Use the pinned normalization and scoring function in `audit_answer_correction.py`: normalized equality to original gold (closed-answer scope), not teacher bits or diagnostic accuracy. Retain full N; abstention counts incorrect. Usability excludes empty/unknown/invalid/uncertain normalized answers. `unanswerable` is an answer token, not a generic abstention. Selection chooses existing normalized answers; return their original acquired raw-answer provenance if displaying outputs.

Fixed candidate model: one logistic regression

For each unique usable normalized answer in the clean + acquired prefix, create one candidate row. Duplicate answer sources are represented by source membership and vote share rather than duplicate candidate rows. Fixed ten numerical features:

- 1-7. Binary source-membership flags: clean, grounding, blur, crop, brightness, noise, blank.
8. Candidate vote share: number of usable observations with that answer divided by number of usable observations.
9. Number of distinct usable answers divided by seven.
10. Number of actually acquired probes divided by six.

No raw answer identity, question text, image features, model/dataset identity, gold correctness, teacher labels, unacquired results, semantic scores, probabilities, confidence, entropy, margins, confidence shifts, APS sets or old diagnostic latents. Only the acquired prefix is passed to the feature function, not a complete teacher row. Keep grouping/scoring metadata in a separate evaluator.

Training label is candidate equality to original gold, used only in fitting. Include B=1,2,3,5,6 prefixes for eligible original-training instances. Weight each candidate by $1 / (\text{number of model records in its group} \times 5 \times \text{number of candidates at that prefix})$; normalize all candidate weights to mean one. Thus groups and budgets receive equal total weight. Skip no-candidate prefixes (record their count); they always abstain at deployment. No class reweighting or oversampling.

Fit one L2 logistic regression: C=1, intercept enabled, lbfgs, tolerance 1e-6, max_iter=2000, random_state=42. Features already bounded; no learned scaling. Pin Python/NumPy/scikit-learn versions in the run manifest before fitting. If solver fails to converge, stop and record failure; do not search architectures or solvers on evaluation outcomes. The exception from the original three-seed neural-net practice is explicit: this is one deterministic convex fit, not a stochastic diagnostic network.

One validation choice

At B=2 only, choose a switch margin from {0, 0.05, 0.10, 0.20, 1.01}. For a usable clean answer, select the highest-scored alternative only if its score is at least 0.5 AND it exceeds the clean score by strictly more than the margin. Otherwise keep clean. Score ties among alternatives select earliest acquisition origin. A tie with clean always keeps clean. If clean is unusable, choose highest-scored usable candidate with earliest-origin tie breaking; if none exists, abstain. Logistic scores are ranking scores, not calibrated confidence claims.

Choose margin maximizing the equal-weight average of the six dataset/model accuracy cells on eligible validation, subject to pooled BreakRate $\leq 1\%$ (damages/originally-correct). Ties: fewer damages, then larger margin. If no margin satisfies the constraint, stop (do not relax it). No other hyperparameter selection. Freeze this same model/margin at all secondary budgets.

Select one baseline comparator on the same validation cells from keep-original, always-ground, majority, supported-ground-else-keep. Highest equal-weight cell accuracy wins; ties prioritize keep, majority, supported-ground, ground. Baseline definitions and ties are exactly those in the audit. Record the chosen comparator and margin before selector evaluation. Calibration is unused.

Baselines and bounds

At each budget report keep-original-matched, always-ground, plurality with clean-preferred ties, supported-ground, and the frozen selector. Also show operational keep-original at cost zero and standalone grounding at cost one separately. Do not call the full-candidate majority cost-matched to a smaller prefix. Report the gold-privileged available-candidate oracle in a separate bound table, paying the same complete prefix; it is not a runnable correction policy. No hindsight choice of cheapest successful probe.

For each split/dataset/model and pooled cohort report N, final-answer accuracy, original accuracy, repairs, damaged correct answers, FixRate, BreakRate, abstentions, mean additional logical cost. Record candidate coverage. Cached latency is supplemental, explicitly excluding failed-attempt history.

Statistical assessment

Primary endpoint: equal-weight mean accuracy across the six closed core-test dataset/model cells at B=2. Also report sample-weighted pooled accuracy and every cell, so composition cannot hide a loss. Secondary budgets and shift are descriptive; no selection or main significance claim from the best secondary result.

Paired cluster bootstrap: 5,000 resamples, seed 20260916; resample canonical image clusters with all questions, models and prefixes together. Cluster draws are global across datasets to preserve any recognized cross-dataset shared image. Recompute the six-cell macro endpoint in each draw; redraw any draw missing a required cell. Report percentile 95% CIs for paired accuracy differences, repair/break rates and pooled accuracy. State limitations of metadata clustering and the exploratory history.

Two primary paired tests: selector vs keep and selector vs validation-selected baseline (deduplicate if baseline is keep). Use 10,000 cluster-level paired randomization draws, seed 20260917: swap the two methods' correctness outcomes for all records in each image cluster with probability 1/2; recompute the primary difference. One-sided $p = (1 + \text{number of permuted differences} \geq \text{observed}) / (10001)$. Holm-adjust the two p-values. Do not treat question/model pairs as independent observations. No significance testing of privileged oracle performance is needed.

Go/no-go, defined before fitting/evaluation

Validation gate: the chosen selector margin must improve six-cell mean accuracy over the validation-selected baseline by ≥ 1 absolute percentage point, with pooled BreakRate $\leq 1\%$. If it fails, stop and report the result; no further selector search.

Exploratory evaluation gate: primary test gain ≥ 1 absolute percentage point over the frozen comparator; the paired CI lower bounds must exceed zero and Holm-adjusted $p < 0.05$ against both keep and comparator; pooled BreakRate $\leq 1\%$; and no dataset/model cell loses > 2 absolute percentage points against keep. Failure means no broad correction-method success claim and no retuning on test/shift. Secondary shift can reveal limits but cannot rescue a failed primary gate. Report all attempted outcomes. These gates do not guarantee publication strength or a particular ICLR score.

Decisions that must be recorded before execution

- Authorize correction as secondary work, or explicitly approve any change to the paper's main objective from frozen behavioral diagnosis.
- Accept closed-answer primary scope, metadata image purge, unchanged original split IDs, and exploratory evaluation status.
- Freeze acquisition order/budgets and the separation from ProActive's existing learned policy.
- Freeze confidence exclusions, ten features, one deterministic logistic fit and sole validation margin choice.
- Freeze normalization, invalid/abstention handling, tie rules and proxy/free-form reporting limits.

- Accept effective logical costs and disclose grounding recovery; do not relabel these as complete historical inference costs.
- Freeze primary endpoint, comparator selection, grouped statistics and go/no-go rules.

The present audit proposes these decisions without writing to the repository's DECISIONS.md. After authorization, all work above can use the existing cached answers; no new large-model inference is needed. Stop here: no selector has been trained.

Appendix B - Audit supplements

Supplement A - baseline definitions copied verbatim from the original audit report. These supplement the specification's cross-reference to "baseline definitions and ties ... in the audit."

Baselines, acquisition, abstention and cost

New descriptive schedule: grounding -> blur -> crop -> brightness -> noise -> blank. Budgets are 0, 1, 2, 3, 5 and 6 additional probes. This schedule is a proposed correction protocol, not a recovered claim about the original policy. All methods at a given positive budget receive and pay for exactly the same prefix. Clean is free under the requested additional-cost convention, but its underlying inference is not literally free.

Usable means valid, applicable, nonempty normalized string other than unknown, invalid, uncertain. Invalid acquired answers cost a probe but do not vote. Literal unanswerable remains a legitimate benchmark answer. Abstention is no usable selected answer, represented as `__ABSTAIN__`; it counts incorrect in the full-cohort denominator. No selective-accuracy filtering.

- Keep-original at B=0 acquires nothing. Keep-original-matched pays the shared B probes and ignores them: this isolates the effect of answer selection.
- Always-ground picks usable acquired grounding, otherwise clean. Its standalone operational cost is one probe. At larger budgets it is an intentionally evidence-matched control; grounding itself does not require the extra probes.
- Majority is plurality over one vote per usable clean/acquired probe answer. Ties keep clean if tied; otherwise choose the earliest tied answer in clean then acquisition order. Repeated answers receive repeated votes.
- Supported-ground-else-keep switches to ground only if another acquired non-clean, non-ground probe agrees with it; otherwise keeps clean. No learned parameter.
- Privileged oracle keeps clean if correct, otherwise chooses the first correct acquired alternative; if none is correct, keeps clean. It uses gold solely to establish a bound. It pays the complete fixed prefix, not a free hindsight-selected single probe.

All mandatory slots are applicable in the audited cohort. For future unavailable/not-applicable slots, skip the call and candidate, charge zero, do not replace it with a later slot; report actual acquired count. An attempted call with an invalid answer costs one. This future behavior is a specification, not empirically exercised by these cache rows.

Costs below are effective logical probe slots. Thirty-nine grounding records were recovered (nine core, thirty shift). Retained latencies omit failed attempts/refresh history. Cached seconds are included as a descriptive proxy in CSV, not end-to-end timing or retry-inclusive cost. Recovery provenance and altered prompt history are potential transfer confounds; no claim of uniform historical physical-call cost is justified. Oracle candidate coverage is only over cached answers, not hypothetical fresh generation.

Supplement B - split counts from the existing cohort `t_index.csv`. These dataset-level counts were not enumerated in the original specification. I have now aggregated the saved flags; this is counting existing audit rows, not a new experiment.

All counts below are **model-example pairs across both models**. Each model has exactly half the displayed count.

Split	Dataset	Original cached pairs	Closed-answer pairs before image exclusion	Closed-answer pairs eligible for fitting/validation
Train	HallusionBench	1,354	1,330	692
Train	POPE	4,168	4,168	780
Train	VizWiz	4,108	0	0
Train	VSR	488	488	426
Train total		10,118	5,986	1,898
Validation	HallusionBench	182	180	116
Validation	POPE	538	538	140
Validation	VizWiz	626	0	0
Validation	VSR	58	58	58
Validation total		1,404	776	314
Calibration	HallusionBench	184	182	Not used
Calibration	POPE	624	624	Not used
Calibration	VizWiz	620	0	Not used
Calibration	VSR	72	72	Not used
Calibration total		1,500	878	Not used
Test	HallusionBench	182	182	Evaluation only
Test	POPE	670	670	Evaluation only
Test	VizWiz	646	0	Excluded from primary scope
Test	VSR	62	62	Evaluation only
Test total		1,560	914	Evaluation only
Shift	IllusionBench	1,200	1,200	Evaluation only
Shift	PRE-HAL	1,200	1,200	Evaluation only
Shift total		2,400	2,400	Evaluation only

The saved audit script defines image identity as follows; this is **actual audit code**, not a new proposed deduplication method:

```

def image_key(t):
    # Known COCO numeric filenames share a namespace across POPE/VSR/PREHAL.
    # This is metadata identity only, NOT image-byte/perceptual deduplication.
    path=t['image_path'];name=Path(path).name
    hit=re.search(r'(?:(?:COCO(?:train|val)\d+_)?(?:(?:jpg|png|jpeg)$)',name,re.I)
    if hit and (t['dataset'] in ('pope','vsr') or 'coco' in path.lower()):return
    'coco:'+hit.group(1)
    if t['dataset']=='hallusionbench':return 'hallusionbench:'+path.split('hallusion_bench/')[1]
    if t['dataset']=='prehal':return 'prehal:'+path.split('/images/')[1]
    return t['dataset']+':'+name

```

The exclusion predicates in that script are:

```

excluded = (
    s == 'train' and bool(other & {'val','cal','test','shift'})
) or (
    s == 'val' and bool(other & {'cal','test','shift'})
)

proposed_fit_eligible = int(
    s in ('train','val') and not excluded
)

closed_answer_eligible = int(
    t['dataset'] != 'vizwiz'
    and t.get('answer_type') != 'open_ended'
)

```

Here other is the set of original splits containing that canonical image identity. Original split assignments are not changed.

Appendix C - Pinned input manifest

These ten file hashes were supplied by the source audit. They identify source inputs; they have not been independently rehashed on this machine. Paths are relative to ProActive/ at commit 14768af3de4436e95238274780efaadbbf8e01db.

```
{
  "inputs": [
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_gemma4_e4b_all_all_shard00-of-04.jsonl",
      "sha256": "46593391081546c310991894934b1436ecfcf1f07956104d59e9e8f4495e1948",
      "rows": 1836,
      "bytes": 15205575
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_gemma4_e4b_all_all_shard01-of-04.jsonl",
      "sha256": "3c7d3ca72753bd49d8f344dd9e468877e75567132de20a81ce20467d52e096a9",
      "rows": 1787,
      "bytes": 14770984
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_gemma4_e4b_all_all_shard02-of-04.jsonl",
      "sha256": "bfff419cdd17e66ccd9eea1a83e248d5f58e16143d662ebd3ba33625b953018ec",
      "rows": 1826,
      "bytes": 15156691
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_gemma4_e4b_all_all_shard03-of-04.jsonl",
      "sha256": "9538751d71a3f8a133be26c2de6d4a4b505b3c780a20ee10a15cd24cd0d8a84e",
      "rows": 1842,
      "bytes": 15271608
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_qwen3_vl_8b_all_all_shard00-of-04.jsonl",
      "sha256": "53cc79788e76a9ddcf8df20812e5ca0ca7aa918dad85d249107f3b46003c5be4",
      "rows": 1822,
      "bytes": 14426686
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_qwen3_vl_8b_all_all_shard01-of-04.jsonl",
      "sha256": "5bcb3927cfca370cce87c95df969dde3f24c26cd901c9733b63bb17180717c65",
      "rows": 1821,
      "bytes": 14398770
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_qwen3_vl_8b_all_all_shard02-of-04.jsonl",
      "sha256": "0ae1afb49f1cf578282a62fafea38ffad93bc7d3220929469d31150e2f9c8711",
      "rows": 1801,
      "bytes": 14265783
    },
    {
      "path":
"outputs/teacher_core_contract_v1_recovered/teacher_qwen3_vl_8b_all_all_shard03-of-04.jsonl",
      "sha256": "4dc9b657f4ea35d7e74e352157f25aee5bd1b2c9c99e13d33755f8f65d55031f",
      "rows": 1847,
      "bytes": 14597023
    },
    {
      "path": "outputs/week8_teacher_recovered/teacher_qwen3_vl_8b_all_shift_shard00-of-01.jsonl",
      "sha256": "38d34e4bc84869126b5dc13355a7697ed0be5a45899ce4049a1bc2ddd0d8601b",
      "rows": 1200,
      "bytes": 8645469
    }
  ]
}
```

```

    "path": "outputs/week8_teacher_recovered/teacher_gemma4_e4b_all_shift_shard00-of-01.jsonl",
    "sha256": "6896756ffe399c99d5702747e54545c49209c5dfe76eaff8052c5f8ed991ea21",
    "rows": 1200,
    "bytes": 8813840
  }
}
}

```

Appendix D - Bundle manifest, reproduction and unresolved settings

Supplement D - complete original ARTIFACT_SHA256 .json. These are SHA-256 file-content hashes, not Git blob IDs. Paths are relative to the extracted bundle.

```

{
  "Cache_Readiness_Report.md": "25275903c228063490ef0a31dde8ef08a9886c517d0bf757df3c391c358462ec",
  "Proposed_Experiment_Specification.md":
    "4b32e10fc902622fa9425dab3410df871661c71428a5adb569d823ca098dfed2",
  "README.md": "078b387d91634cfd2f412d3922743acc8e6e16fad328df5f44e9d3affcfa2c2",
  "REPORT_TEMPLATE.md": "0badb61b126deaace8b023c4e147f00c3802057268c08d5b999c03474c7ed461",
  "audit_answer_correction.py":
    "022350fc776514252cc696f2ab2a72297bc02aaad8853c5f0418bbe4c4a41c1a",
  "build_report.py": "f7c35a8380e330eda855fec23de88b459d863c1cfa82df38e149992045624274",
  "reference_results/audit_summary.json":
    "d6e8f432e15594729d764facbf0c600e6d45559fe1b971ce5c14b9ac43e82d38",
  "reference_results/baseline_instances.csv":
    "9f8c8b9de53c0bbd20725f248cbd2b67087cfce3d3d4beb5f557f9071b311444",
  "reference_results/baseline_tables.csv":
    "5590a85163ef04d1ea7310788b207d5aa9e27416892904250ee7afd745c852fd",
  "reference_results/cache_inventory.csv":
    "a04d2d5bf515e59ec445fa9e6216d347f092bc368c466ccf7ac4571f72b34583",
  "reference_results/closed_answer_summary.csv":
    "e1a874971ed7eeb3275e853700d4efe110c28741e724e8f60d7538c346c61827",
  "reference_results/cohort_index.csv":
    "a4cad1636cf11c5cc78ef580de189f0356d9453e96814bd2bcf3171456b45cfe",
  "reference_results/image_question_split_overlap.csv":
    "3dc9d701f439f9ad19733f02b15904cc116084be1edb308c7c32f8db4bed0562",
  "reference_results/image_split_overlap.csv":
    "2f51182e6ef57f0fb00650af91c46332056bd018ccb053abeb93809b2181c00e",
  "reference_results/proposed_fit_counts.csv":
    "aba66fadf9575724614069faee5160810a1a424d3a4702ca7a3e7d16cc1d26f1",
  "reference_results/recovered_groundings.csv":
    "340ba49bfe4969d94722abe98103a21a8966150ab5f05b9bcc039ea313daf903",
  "reference_results/unusable_probe_answers.csv":
    "6438a18773576c148862cb1a7c941d144dec427a1c18cedfcfffb32c45b15b0c9",
  "source/AGENTS.md": "c8e38935e52c4e07f8a5f52173193f424418a84c10ef9f88de158c0704e14828",
  "source/DECISIONS.md": "d4e6053abf4956e7342cc91244f66b01a0c2264629f77f1d98f3448f707b0650",
  "source/PROACTIVE_RESEARCH_HANDOFF_AND_RESULTS_INDEX.md":
    "2a811ee073c873e8e752551c4263a294517404f0e16b40f24eb0871409b1870d",
  "source/PROJECT_STATUS.md": "f316a405cf41eda9978e23dc6c9955f380af9f0f3d5743a7c97f569f96293843",
  "source/configs/experiments/teacher_core.yaml":
    "9820cf2c566ddad75c218842cb410813ca8ab9c8a7fa958d2053bf12958f5481",
  "source/instructions.md": "3fc2cb7c11b60209134ea97e143e5dae858b40465ff0854e555c949a1839bbe0",
  "source/outputs/week4_reports/final/teacher_manifest.json":
    "5f04d6340ba36dd340b0071ae2236324a757691519028f79d74388780a867029",
  "source/src/proactive/data/hallusion_contract.py":
    "b8a9e6c91f43cc82e756710d6714769da0d48c1552cc049fbcb3dd449bb39310",
  "source/src/proactive/data/splits.py":
    "9fbf8f7a0a369f69e1f4a07072e20afa566ea64072dd9e564dc024a4c5d6eafa",
  "source/src/proactive/features/normalization.py":
    "81c377977f84182ce91da4e1ad81ba532cf0e351e91e12c6cc8a4f813fd13991",
  "source/src/proactive/features/semantic.py":
    "9ab90c329e3cbab267a25b54d7ccc008e5156ebd9cd0f1f0debbb5b4c03aebb0",
  "source/src/proactive/probes/probe_runner.py":
    "413d6eaf4c22a95fdba7c6e5546f4622494307971dd0c676c7d09e9f296f5f2f",
  "source/src/proactive/teacher/cache_builder.py":
    "e980c1d9e3fd5b44f9822cd4d8049efed2b873449cf442002b7cf4f449b9cb72",
  "source/src/proactive/teacher/offline.py":
    "7b2f3247fbb917ef1a889ba6bfce81982e76cb44ff115fc995c8769019595084",
  "source/v3.5_ProActive_Complete_Super_Implementation_Plan.md":
    "4c9fd55fb77f06c9aa27c32ae35700a97ef0131b845b91a423940d0575d94d78",
  "source_provenance.json": "c719c608f05a51b86f68f29a5567ed6c2e4387b530d070dc17bc5b846ebd8c10"
}

```

```
}
```

The ZIP's SHA-256, checked during the preceding retrieval, is:

```
ProActive_PlanB_Audit_Bundle.zip  
a26a21ce2e6f73a5db96874a2cd0fde64aa8297048e1b8019685f16986e9b727
```

Reproduction commands from the original README:

```
python audit_answer_correction.py --download  
python build_report.py
```

Run from a **fresh extraction** of the bundle, using Python 3.11+. These commands reproduce the audit only; they do not train the proposed selector.

Requirements:

- Standard library only for audit reproduction.
- Network access to download the ten pinned input caches.
- Approximately 135 MB of raw cache inputs.

Directory behavior:

- `reference_results/` contains the packaged, saved audit outputs.
- `results/` receives newly reproduced outputs.
- `audit_answer_correction.py` refuses an already existing `results/` directory.
- `data/` receives downloaded caches.
- Raw caches are excluded from the ZIP.
- Input SHA-256 hashes are checked before analysis.
- The full baseline table has 1,534 rows.
- Accuracy columns in CSV are fractions, not percentages.

The actual downloader constructs each cache URL as:

```
https://media.githubusercontent.com/media/gitgoap/ProActive-ICLR-  
Dump/14768af3de4436e95238274780efaadbbf8e01db/ProActive/{path}
```

Here `{path}` is an exact input path from the audit summary. This is a **URL template for repository inputs**, not a public URL for the generated specification.

Accessible repository references:

- [Pinned repository directory](#)
- [Pinned handoff and results index](#)
- [Pinned core teacher manifest](#)

No public HTTPS download URL was created for these generated artifacts. These are conversation downloads, not verified public links for external PDF readers.

Unspecified details and implementation cautions - retrieval notes, not additions to the original specification:

1. **Selector score API is not explicitly specified.** The original says "logistic scores," with a 0.5 threshold and switching margins, but does not explicitly state `predict_proba(...)[positive_class]` versus another API. The intended bounded score is suggested by those thresholds, but the student instruction should record the exact definition before implementation.
2. **Exact Python, NumPy and scikit-learn versions are not specified.** The document instructs the implementer to pin them before fitting. No selector environment lockfile or trained model exists in this bundle.
3. **Random-number generator implementation, cluster enumeration order and percentile interpolation method are not specified.** Seeds and resample counts are specified; exact bit-for-bit statistical reproduction needs these additional implementation choices recorded.
4. **Near-equality tolerance is not specified.** Tie rules are specified, but no floating-point tolerance for comparing scores or validation metrics is given.

5. **Handling a bootstrap sample with zero originally wrong or zero originally correct examples is not specified.** The document explicitly handles missing dataset/model cells, but does not define undefined FixRate/BreakRate denominators within a resample.
6. **Duplicate normalized candidates' displayed raw-answer provenance is not fully specified.** The specification asks to retain raw-answer provenance, but does not prescribe which raw string to display when multiple sources normalize to the same selected answer. This does not change normalized-answer scoring.
7. **Future unavailable probes are described in the proposal but not supported as an exercised audit case.** The actual audit script asserts that all six mandatory cache slots are valid and applicable. Its cost is therefore `len(actions)` for the acquired prefix. A general deployment implementation would need to implement the proposal's skip-and-charge-zero behavior explicitly.
8. **The original no-candidate weighting statement needs care.** It says groups and budgets receive equal total weight, while also saying to skip no-candidate prefixes. Equal total weight is not guaranteed for a hypothetical group with skipped prefixes unless further normalization is defined. No such additional normalization is specified.
9. **There is no selector-training or statistical-test executable in the bundle.** `audit_answer_correction.py` computes descriptive cached baselines; `build_report.py` renders the report and closed-answer summaries. Training and statistical procedures are proposed text, not already implemented results.

These points should remain marked as **decisions to record before execution**, rather than silently filled in and attributed to the original artifact.